

Who holds government debt?

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Abstract

Across advanced economies at similar debt-to-GDP ratios, the same stock of government debt is held by sharply different mixes of domestic non-banks, foreigners, and the banking sector. I propose a cost-minimizing model in which a government places a given debt stock across three sectors with different marginal costs: non-banks with a country-specific capacity, foreigners with a common ceiling, and a banking system with no hard cap but convex placement costs. The model predicts that countries with deeper non-bank investor bases absorb more debt at every level, that the bounded sectors saturate concavely, and that banking takes over the margin once total debt is high enough. Using the Arslanalp–Tsuda panel of sovereign debt by holder for 32 advanced economies (1989–2021), a two-stage estimator finds a sharp kink in banking absorption at 102.7% of GDP, with combined central- and private-bank marginal absorption rising by roughly 82 percentage points above the threshold. Non-bank concavity operates within country (capacity); foreign concavity operates across countries (common ceiling).

Keywords Sovereign Debt; Debt Composition; Banking Takeover

JEL codes H63; F34; E62

This paper is distributed as a REMARK-style replication package for the Johns Hopkins Econ 606 course project (Spring 2026): it pairs this PDF with executable code that reproduces the empirical specifications reported in the quantitative sections. **GitHub repo:** <https://github.com/emma0502/emma0502606> (tagged release v0.1.0; indexed in the [econ-ark/REMARK](#) catalog). **Replication code:** Stata in `Code/empirical_debt_composition/do/`; Python companion in `Code/empirical_debt_composition/python/`. Reproduce from the repository root with `./reproduce.sh` (see `README.md`).

1 Introduction

Government debt has risen substantially across advanced economies in recent decades. Yet countries with similar levels of public debt often differ sharply in who holds that debt. In some countries, government debt is absorbed mainly by domestic non-bank investors such as pension funds, insurance companies, and households. In others, a larger share is held by foreign investors or by the domestic banking system, including private banks and the central bank. This raises a central question: when a government has more debt to place, what determines who ends up holding it, and why does this allocation differ across countries?

This question matters for both fiscal and financial stability. The creditor composition of public debt affects the strength of the sovereign–bank nexus, the extent of fiscal space, and the economy’s exposure to global investor sentiment. A government that relies heavily on foreign investors may be more vulnerable to shifts in global risk appetite, while a government that places a large share of debt with domestic banks may deepen the connection between sovereign risk and financial-sector fragility. Similarly, a country with a large domestic non-bank investor base may be able to absorb public debt with less pressure on the banking system or foreign funding.

Despite its importance, the allocation of sovereign debt across creditor sectors remains insufficiently understood. Existing work often studies one creditor sector at a time, or documents differences in debt-holder composition without providing a joint account of why the split across sectors takes the form it does. What is missing is a framework that explains how a given stock of government debt is allocated across domestic non-banks, foreign investors, and the banking sector, and how that allocation depends on both the level of debt and the country’s domestic financial structure.

This paper addresses that gap by developing a cost-minimizing model of sovereign debt allocation. The government must place a given amount of debt and chooses how much to allocate to three sectors: domestic non-banks, foreign investors, and the banking sector. Each sector differs in its effective capacity and marginal placement cost. Domestic non-bank capacity varies across countries with the depth of the local financial system. Foreign demand is limited by a common exposure ceiling. The banking sector has no hard ceiling, but greater reliance on banks carries increasing quasi-fiscal and financial-stability costs. The model therefore links cross-country differences in debt-holder composition to differences in domestic financial structure and predicts how marginal debt absorption changes as total debt rises.

We structurally estimate the model using panel data on debt composition across countries and over time. The estimation identifies distinct borrowing limits across creditor types and provides a tractable framework to interpret variation in who holds government debt.

2 Motivation: two stylized facts

Two patterns in the cross-country data on advanced-economy sovereign debt motivate the framework that follows. First, a sharp *kink* at high debt: once total government debt crosses roughly 100% of GDP, banking-sector holdings rise much more steeply than below the threshold. Second, substantial *cross-country heterogeneity* at any given debt level: countries with similar debt-to-GDP ratios place their debt in very different hands.

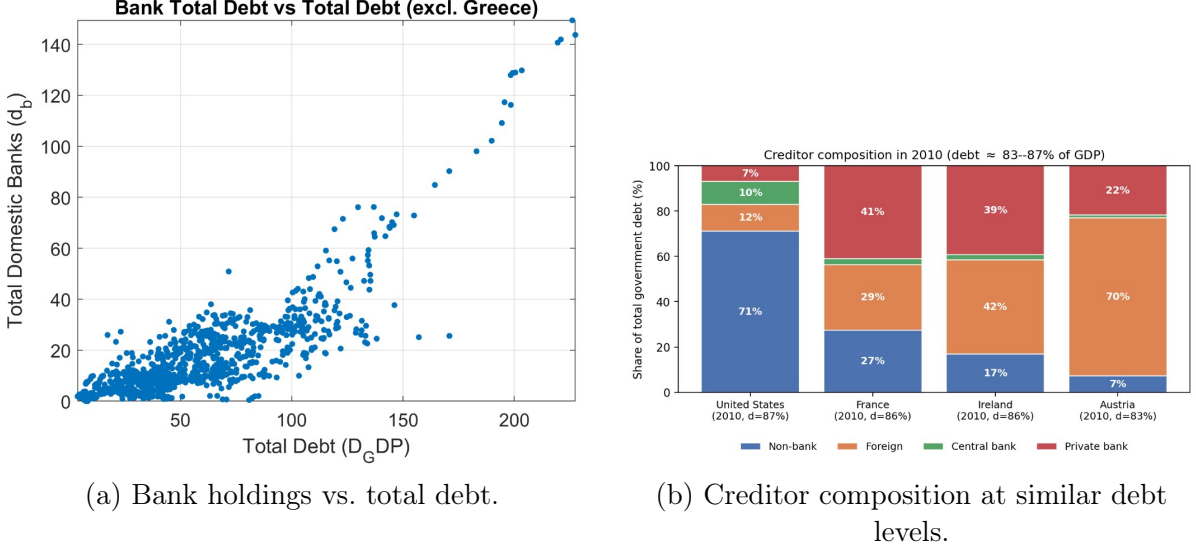


Figure 1 Motivating evidence.

2.1 Fact 1 — Banking takeover above a high-debt threshold

Once total government debt rises beyond roughly 100% of GDP, the banking sector begins to absorb a noticeably larger share of marginal debt issuance than at lower debt levels. Below the threshold, an additional unit of total debt translates into only a modest increase in combined central-bank and private-bank holdings; above it, the same incremental unit of debt is absorbed mostly by the banking system. The slope of bank holdings against total debt steepens sharply around this point, and banking eventually accounts for nearly all of the marginal issuance. This is the empirical pattern we will refer to as the “banking takeover” and which the model in Section 4 is designed to rationalize.

2.2 Fact 2 — Cross-country heterogeneity at the same debt level

At any given debt-to-GDP ratio, advanced economies place their debt with strikingly different creditor sectors. Some lean heavily on domestic non-bank investors such as pension funds and insurance companies; others rely on foreign investors; others place a large share of their debt with the domestic banking system. This heterogeneity does not

vanish once we condition on the level of debt: countries with intrinsically larger non-bank financial sectors absorb a larger share of government debt through non-banks, while countries with structurally smaller non-bank sectors push more debt onto foreigners and the banking system.

2.3 What this implies for a framework

Together the two facts pin down what a useful framework needs to deliver. It must (i) generate a sharp threshold in marginal banking absorption as total debt rises, and (ii) allow the cross-country mix of creditors at a given debt level to vary with the country's domestic financial structure. Section 4 develops a cost-minimization model that produces both features: domestic non-banks have country-specific capacity, foreign investors face a common exposure ceiling, and banking is the residual margin with no hard ceiling but a convex placement cost. Section 5 takes that framework directly to panel data and shows that both stylized facts emerge as predicted.

3 Related literature and organization

3.1 Related literature

This paper speaks to three related strands of literature: theoretical work on sovereign debt demand, empirical work on the holder composition of sovereign debt, and the literature on high-debt regimes and financial repression. The central gap across these strands is that existing work typically studies either one creditor sector at a time or documents creditor composition descriptively. By contrast, this paper develops a joint framework in which a government allocates a given stock of debt across domestic non-banks, foreign investors, and the banking sector, and then brings the model directly to data.

3.1.1 *Sovereign debt demand and creditor-sector mechanisms*

A first strand of literature studies why particular sectors hold government debt. In models of domestic non-bank or household demand, government debt can provide liquidity and self-insurance services. For example, Aiyagari and McGrattan (1998) analyze the optimal quantity of public debt in an incomplete-markets environment, where government bonds help households smooth consumption in the presence of idiosyncratic risk. This mechanism highlights why domestic private agents may willingly absorb government debt even when debt is not needed to finance current spending.

A second branch focuses on foreign demand for sovereign debt, especially in the presence of default risk. In canonical sovereign-default models such as Arellano (2008), foreign investors price sovereign debt according to the government's endogenous default incentives. Borrowing is limited not by an exogenous portfolio capacity, but by the interaction between default risk, interest rates, and the government's willingness to repay. This literature is central for understanding external debt and international investor

discipline, but it does not ask how foreign demand competes with domestic non-bank or bank demand for a common stock of government liabilities.

A third branch studies banks as sovereign-debt holders. The sovereign–bank nexus literature emphasizes that banks may become large holders of domestic government debt because of regulation, collateral needs, implicit guarantees, or financial repression. Brunnermeier, Garicano, Lane, Pagano, Reis, Santos, Thesmar, Van Nieuwerburgh, and Vayanos (2016), for example, describe the sovereign–bank “diabolic loop,” in which weak sovereigns and fragile banks reinforce each other through banks’ holdings of domestic public debt. This literature explains why bank holdings are macro-financially important, but it generally treats the banking sector in isolation rather than as one sector among several competing buyers of government debt.

The limitation of these theoretical approaches is therefore not that they miss important mechanisms. Rather, each mechanism is usually developed separately. Households and non-banks provide liquidity demand, foreigners price default risk, and banks generate sovereign–bank feedbacks. What is missing is a joint allocation problem: given a fixed amount of government debt, how much is placed with each sector, and how does that allocation change with the level of debt and with the country’s financial structure? This paper addresses that gap by placing the three creditor sectors in a single cost-minimization framework.

3.1.2 Empirical evidence on the composition of sovereign debt

A second strand of literature documents who holds sovereign debt across countries and over time. Arslanalp and Tsuda (2014) provide a key dataset for advanced-economy sovereign debt holdings by creditor sector. Their work makes it possible to study the evolution of sovereign debt demand across domestic banks, central banks, non-banks, and foreign investors in a systematic cross-country panel.

More recent empirical work emphasizes that the creditor composition of sovereign debt matters for macroeconomic outcomes. Fang, Hardy, and Lewis (2025) show that the identity of sovereign-debt holders varies meaningfully across countries and is related to fiscal and financial vulnerabilities. This work is closest to the empirical motivation of the present paper because it focuses not only on the level of public debt, but also on who holds that debt.

However, the existing empirical literature is primarily descriptive. It documents that creditor mixes differ across countries and that the composition of holders changes as debt rises, but it does not provide a structural account of why the split across sectors takes the form it does. In particular, it does not explain why, at similar debt-to-GDP levels, some countries place debt mainly with domestic non-banks, others with foreigners, and others with the banking sector. Nor does it explicitly connect cross-country heterogeneity in debt composition to differences in domestic financial structure. This paper uses the Arslanalp–Tsuda debt-holder panel, but interprets the data through a model in which sectoral absorption depends on creditor-sector capacity and marginal placement costs.

3.1.3 High debt, thresholds, and financial repression

A third related literature studies what happens when public debt becomes very high. One influential line of work focuses on debt thresholds and the macroeconomic consequences of high debt. Reinhart and Rogoff (2010) argue that high public debt is associated with weaker growth, motivating a broad literature on whether debt becomes especially costly above certain levels.

More directly related to this paper is the literature on financial repression and the role of banks as marginal buyers at high debt levels. Jeanne (2025) develops a framework in which, once fiscal adjustment becomes politically or economically difficult, governments may increasingly rely on the domestic financial system to absorb public debt. This mechanism is closely connected to the high-debt pattern studied here: above a threshold level of debt, the banking sector becomes the dominant marginal absorber of additional government liabilities.

The contribution of the present paper is to embed this high-debt banking takeover into a broader allocation framework. The high-debt threshold is only one part of the debt-allocation problem. Below the threshold, most of the cross-country variation lies in the split between domestic non-banks and foreign investors. This paper therefore adds a below-threshold mechanism: domestic non-bank absorption depends on country-specific sector size, while foreign absorption is disciplined by a common foreign-investor ceiling. The banking sector then becomes increasingly important only when the other sectors approach their effective limits.

3.1.4 Contribution of this paper

Relative to the theoretical literature, the paper’s first contribution is to build a joint three-sector model of sovereign debt allocation. A government must place a given amount of debt and chooses the allocation across domestic non-banks, foreign investors, and the banking sector to minimize total placement cost. Domestic non-banks have a country-specific capacity, foreign investors face a common exposure ceiling, and the banking sector has no hard ceiling but carries an increasing quasi-fiscal cost. The model therefore generates predictions for both the level and marginal composition of sovereign debt holdings.

Relative to the empirical literature, the paper’s second contribution is to connect the model directly to cross-country data. The empirical strategy uses variation in sectoral financial structure to explain why countries with similar debt levels have different creditor mixes. Country-level structural sector sizes are recovered from financial-sector data and then used to explain sectoral debt absorption in the Arslanalp–Tsuda sovereign-debt-holder panel. This provides a structural interpretation of patterns that the existing empirical literature has mostly documented descriptively.

Finally, relative to the high-debt and financial-repression literature, the paper’s third contribution is to show that banking-sector absorption is part of a broader pecking-order logic. Banking takeover occurs not simply because banks are always the preferred buyers, but because domestic non-banks and foreign investors become increasingly saturated as

debt rises. The model and empirical results therefore connect the high-debt banking threshold to the lower-debt allocation between non-bank and foreign creditors, producing a unified account of sovereign debt composition across the full debt distribution.

3.2 Organization

The paper is organized as follows. Section 2 documents two stylized facts that motivate the framework: a banking-takeover kink at high debt and cross-country heterogeneity in creditor mix at any given debt level. Section 4 presents the three-sector cost-minimization model, in which the government allocates debt across domestic non-banks with country-specific capacity, foreigners with a uniform no-default ceiling, and banks as the residual absorber with convex financial-repression costs, and derives the resulting equilibrium predictions for debt composition. Section 5 takes the model to the data: we describe the panel of advanced-economy debt-composition data and structural sector-size variables, implement a two-stage estimator that identifies structural sector sizes in a first stage and the cross-country allocation of debt in a second stage, and document the resulting heterogeneous pecking order and concavity in debt absorption. Section 6 concludes.

4 Model

A cost-minimising government chooses how to allocate a given stock of debt across three creditor sectors — domestic non-banks, foreigners, and the domestic banking system (central bank plus private banks, treated as one block). The three sectors differ in *how* placement becomes costly at the margin, and those differences produce a tractable pecking order with sharp comparative-static predictions that we take to the data.

4.1 Setting

The government has a given stock of outstanding debt $D > 0$ and chooses non-negative holdings (d_h, d_f, d_b) satisfying the budget constraint

$$d_h + d_f + d_b = D, \tag{1}$$

where d_h is non-bank holdings, d_f is foreign holdings, and d_b is banking absorption. Each sector is summarised by the cost of placing debt with it. Non-banks and foreigners lend at interest rates $r_h(d_h, S_h)$ and $r_f(d_f)$, so the government's cash cost of placing d_h and d_f is the interest bill $r_h d_h + r_f d_f$. The non-bank schedule depends on a country-specific structural size $S_h > 0$, which captures the depth of the domestic non-bank investor base. The banking sector is not described by an interest rate: placing d_b with the banking system carries a separate welfare cost $\Psi(d_b)$ that proxies for inflation, monetary–fiscal trade-offs, and financial-stability risk. The planner therefore chooses $d_h, d_f, d_b \geq 0$ to

minimise total economic cost subject to the budget constraint:

$$\min_{d_h, d_f, d_b} r_h(d_h, S_h) d_h + r_f(d_f) d_f + \Psi(d_b) \quad \text{s.t.} \quad d_h + d_f + d_b = D. \quad (2)$$

4.2 The three sectors

Marginal cost rises with the sector's degree of saturation. What differs is *how* saturation is measured.

Non-banks. Non-banks operate in a segmented domestic market. Their interest rate depends on the portfolio share d_h/S_h , not on d_h alone: the same absolute amount of debt is harder to place in a country with a thinner non-bank investor base. We adopt the schedule

$$r_h(d_h, S_h) = \bar{r}_h + \gamma_h \frac{d_h/S_h}{\bar{\omega} - d_h/S_h}, \quad 0 \leq d_h/S_h < \bar{\omega}, \quad (3)$$

with $\bar{r}_h, \gamma_h, \bar{\omega} > 0$. The rate stays near $\bar{r}_h$ when the sector is far from its country-specific cap $\bar{\omega}S_h$ and diverges as $d_h \rightarrow \bar{\omega}S_h$. A larger S_h shifts the capacity outward, so the same d_h leaves more headroom and is cheaper at the margin. Cross-country differences in S_h are the empirical hook of the model and the structural object recovered by Stage 1 of the two-stage estimator in Section 5.

Foreigners. Foreign participation is disciplined by a no-default / incentive constraint that imposes a uniform upper bound $\bar{D}_f$ on foreign holdings, common across countries. Risk premia rise as the sovereign approaches that ceiling. We posit

$$r_f(d_f) = \bar{r}_f + \gamma_f \frac{d_f}{\bar{D}_f - d_f}, \quad 0 \leq d_f < \bar{D}_f, \quad (4)$$

with $\bar{r}_f, \gamma_f > 0$. The schedule is flat when $d_f \ll \bar{D}_f$ (the global market is deep) and diverges as $d_f \rightarrow \bar{D}_f$.

Banking. The banking sector absorbs the residual $d_b = D - d_h - d_f$. Unlike non-banks and foreigners, it has no hard ceiling — the central bank can in principle keep expanding its balance sheet — but doing so is distortionary. We summarise that distortion by an increasing, convex welfare cost

$$\Psi(d_b), \quad \Psi'(d_b) > 0, \quad \Psi''(d_b) > 0. \quad (5)$$

The level $\Psi'(0)$ is the cost of placing the first dollar with banks and is directly comparable to $\bar{r}_h$ and $\bar{r}_f$. For the numerical illustrations below we use the simplest affine-marginal-cost specification

$$\Psi(d_b) = \bar{r}_b d_b + \frac{\psi}{2} d_b^2, \quad \Psi'(d_b) = \bar{r}_b + \psi d_b, \quad \psi > 0, \quad (6)$$

with $\bar{r}_b, \psi > 0$.

4.3 Equilibrium and closed-form solution

With Lagrange multiplier λ on the budget constraint, the interior first-order conditions equate marginal placement costs across all three sectors:

$$\text{MC}_h(d_h, S_h) = \text{MC}_f(d_f) = \Psi'(d_b) = \lambda(D), \quad (7)$$

where, from (3) and (4),

$$\text{MC}_h(d_h, S_h) = \bar{r}_h + \gamma_h \frac{(d_h/S_h)[2\bar{\omega} - d_h/S_h]}{[\bar{\omega} - d_h/S_h]^2}, \quad (8)$$

$$\text{MC}_f(d_f) = \bar{r}_f + \gamma_f \frac{d_f(2\bar{D}_f - d_f)}{(\bar{D}_f - d_f)^2}. \quad (9)$$

If any sector were cheaper at the margin, the planner would shift toward it.

The system (1)–(7) admits a closed-form parametric solution in the shadow rate λ :

$$d_h^*(\lambda, S_h) = S_h \bar{\omega} \left(1 - \frac{1}{\sqrt{1 + (\lambda - \bar{r}_h)/\gamma_h}} \right), \quad (10)$$

$$d_f^*(\lambda) = \bar{D}_f \left(1 - \frac{1}{\sqrt{1 + (\lambda - \bar{r}_f)/\gamma_f}} \right), \quad (11)$$

$$d_b^*(\lambda) = \frac{\lambda - \bar{r}_b}{\psi}, \quad (12)$$

with $D(\lambda) = d_h^* + d_f^* + d_b^*$ strictly increasing in λ on the admissible range. Inverting numerically yields the equilibrium allocation as a function of D .

4.4 Comparative statics: who absorbs the next dollar?

The substantive content of the model rests on a single structural property of the three placement-cost schedules: in each sector, the marginal cost of placing additional debt is strictly increasing in that sector's existing holdings. Differentiating (8)–(9) and (6) gives the slopes of the marginal-cost schedules,

$$H_h \equiv \frac{\partial \text{MC}_h}{\partial d_h} = \frac{2\gamma_h \bar{\omega}^2}{S_h [\bar{\omega} - d_h/S_h]^3} > 0, \quad (13)$$

$$H_f \equiv \frac{\partial \text{MC}_f}{\partial d_f} = \frac{2\gamma_f \bar{D}_f^2}{[\bar{D}_f - d_f]^3} > 0, \quad (14)$$

$$H_b \equiv \frac{\partial \text{MC}_b}{\partial d_b} = \psi > 0, \quad (15)$$

with $H_h \rightarrow \infty$ as $d_h \rightarrow \bar{\omega}S_h$ and $H_f \rightarrow \infty$ as $d_f \rightarrow \bar{D}_f$. The banking slope is constant; the other two diverge as their respective sectors saturate.

Total differentiation of the FOC (7) together with the budget constraint (1) delivers the marginal share of new debt absorbed by each sector,

$$s_j \equiv \frac{\partial d_j^*}{\partial D} = \frac{1/H_j}{1/H_h + 1/H_f + 1/H_b}, \quad j \in \{h, f, b\}, \quad (16)$$

which sum to one by construction. The marginal share of each sector is therefore inversely proportional to the local curvature of its marginal cost: at low D all three H_j are bounded and the s_j are roughly constant; as $d_h \rightarrow \bar{\omega}S_h$ and $d_f \rightarrow \bar{D}_f$, H_h and H_f diverge, s_h and s_f collapse to zero, and $s_b \rightarrow 1$.

4.5 Three predictions

The structural property $H_j > 0$ delivers three substantive comparative-static results, which we record as propositions. Proofs are collected in Appendix A.

Proposition 1 (Sectoral holdings rise with total debt). *Under upward-sloping marginal cost in every sector ($H_j > 0$), d_j^* is weakly increasing in D for $j \in \{h, f, b\}$, and strictly increasing whenever $d_j^* > 0$.*

Intuition. If, when the government must place more debt, some sector's holdings actually fell, that sector's marginal cost would drop. The FOC requires all active sectors to share the same marginal cost, so the others would have to drop too — meaning every sector absorbs less even though total debt has risen. That is impossible, so no sector can shrink as D grows.

Proposition 2 (Non-bank absorption rises with non-bank sector size). *Fix total debt $D > 0$ and suppose the equilibrium is interior for non-banks with $d_h^* > 0$. Then $\partial d_h^* / \partial S_h > 0$.*

Intuition. Non-bank marginal cost depends on the portfolio share d_h/S_h , not on d_h alone: the sector becomes expensive only as d_h/S_h approaches $\bar{\omega}$. A larger S_h means that any given level of d_h corresponds to a smaller share, so marginal cost is lower; holding D fixed, the planner tilts the allocation toward non-banks when S_h rises. This is the mechanism behind the cross-country pecking order in Figure 3, and it is the structural object recovered by Stage 2 of the empirical estimator.

Proposition 3 (Strict concavity in the symmetric calibration). *Suppose $\bar{r}_h = \bar{r}_f$, $\gamma_h = \gamma_f$, and the equilibrium path is interior for both capped sectors. Then $d_h^*(D)$ and $d_f^*(D)$ are strictly concave in D on the interior of their domains.*

Intuition. Each sector's holdings rise with total debt, but ever more slowly: placement is cheap when a sector is far from its ceiling and expensive as it approaches $\bar{\omega}S_h$ (non-banks) or $\bar{D}_f$ (foreigners), so marginal issuance tilts toward whichever margin still has room. Both capped curves therefore bend downward, even though the government continuously substitutes across sectors as relative costs shift.

Banking takeover at high debt. The three propositions sharpen into a single qualitative prediction for the high- D region. As $d_h \rightarrow \bar{\omega}S_h$ and $d_f \rightarrow \bar{D}_f$, H_h and H_f diverge while H_b remains constant; (16) then forces $s_h, s_f \rightarrow 0$ and $s_b \rightarrow 1$. Because the non-bank ceiling $\bar{\omega}S_{h,i}$ varies across countries while $\bar{D}_f$ is common, each country has its *own* takeover level $\bar{D}_i^{\text{bt}} \approx \bar{\omega}S_{h,i} + \bar{D}_f$ at which banking begins to absorb the bulk of new issuance. Once D exceeds $\max_i \bar{D}_i^{\text{bt}}$, every country has exhausted its capped sectors and banking absorbs nearly every additional dollar of debt regardless of S_h . Cutting the horizontal axis at $D \approx 100\%$ of GDP therefore gives a uniform picture across countries: this is the theoretical counterpart of the pooled empirical kink we estimate in Section 5.

4.6 Numerical illustration: three phases of absorption

We illustrate the equilibrium with the baseline calibration $\bar{r}_h = \bar{r}_f = \bar{r}_b = 1$, $\gamma_h = \gamma_f = 0.02$, $\bar{\omega} = 0.30$, $\psi = 0.04$, and $\bar{D}_f = 55$. Figure 2 traces the equilibrium for a single country with $S_h = 150$ and shades the three phases defined by banking's marginal share $s_b = \partial d_b^* / \partial D$. In Phase 1 both capped sectors are far from their caps, H_h and H_f are bounded, and (16) delivers a small s_b : non-banks and foreigners absorb most new debt while banking is essentially flat. As D rises into Phase 2, d_h and d_f approach their caps, H_h and H_f rise sharply, and the planner shifts the marginal dollar toward banking. In Phase 3 both capped sectors are close to their limits, $1/H_h$ and $1/H_f$ collapse to zero, and s_b approaches one.

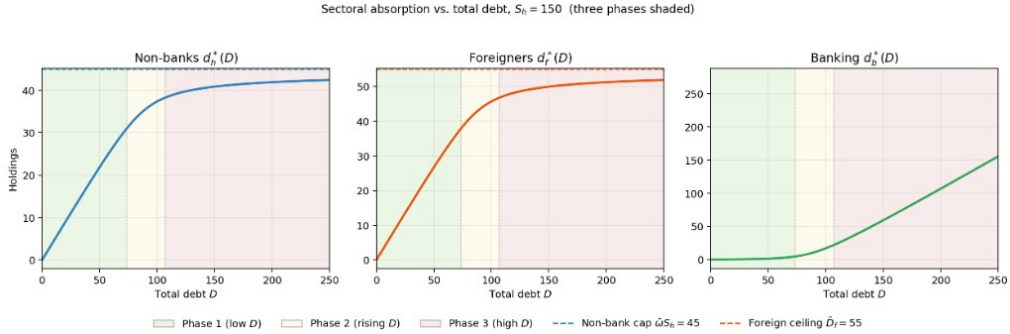


Figure 2 Three phases of debt absorption.

4.7 Numerical illustration: heterogeneous non-bank sector

Figure 3 overlays equilibrium allocations for five countries that differ only in non-bank capacity, $S_h \in \{30, 70, 110, 150, 200\}$, holding $\bar{D}_f = 55$ and all other parameters at the baseline. Because $1/H_h$ in (16) is increasing in S_h , the model implies a pecking order in S_h : at every level of D , countries with smaller non-bank sectors hold a higher share of debt with foreigners and with banks. The middle panel shows that small- S_h economies climb the foreign schedule faster and hit the uniform ceiling $\bar{D}_f$ at lower D ; the right panel shows that they also load more onto banking, and their banking curves

lie uniformly above those of large- S_h countries. Once D is large enough that every country has crossed its own $\bar{D}_i^{\text{bt}}$, the five banking curves become approximately parallel with $s_b \rightarrow 1$, while non-bank and foreign curves are flat at their respective caps — the cross-country signature of the banking-takeover prediction.

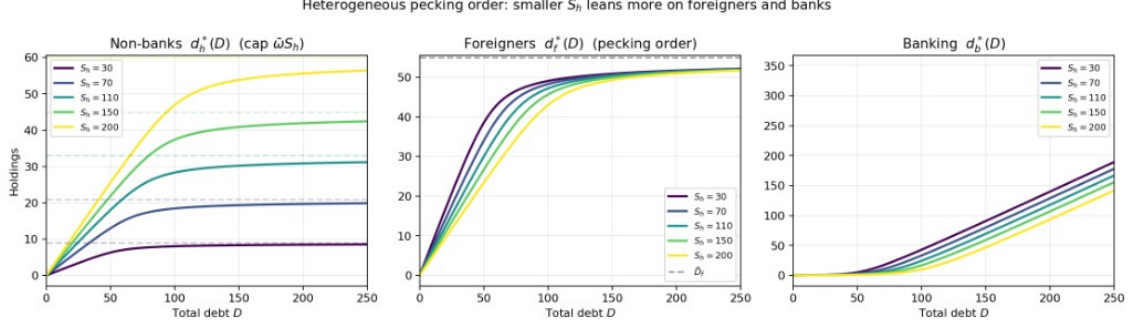


Figure 3 Pecking order across countries with different non-bank sector sizes S_h .

5 Empirical Analysis

We bring the cost-minimizing allocation model of Section 4 to a panel of 32 advanced economies over 1989–2021. The empirical strategy is a two-stage estimator: a first stage recovers each country’s structural sector size $\hat{\delta}_{ij}$, and a second stage uses these as interaction variables to estimate the slope and threshold behaviour of debt allocation.

5.1 Data and sector-size variables

Panel. The unit of observation is a country–year. The sample covers 32 advanced economies over 1989–2021, for a balanced maximum of $32 \times 33 = 1,056$ country–year observations. Coverage is unbalanced: it thickens after ~ 2000 and is full from 2007 onward. Cyprus is dropped from non-bank specifications because the underlying non-bank size variable is missing for all years; the headline sample of $N = 780$ observations refers to the central-bank and private-bank Stage 2 regressions.

Variables. We use two data sources.

- **Debt holdings.** Arslanalp and Tsuda (2014), sovereign debt by creditor sector. We observe holdings by three domestic sectors—non-banks (d_h), central bank (d_{cb}), and private banks (d_{pb}). Foreign holdings d_f are treated as the accounting residual:

$$d_{f,it} = D_{it} - d_{h,it} - d_{cb,it} - d_{pb,it}. \quad (17)$$

- **Sector-size proxies.** World Bank, *Global Financial Development Database* (GFDD). We construct a country–year proxy $s_{j,it}$ for the structural size S_j of each

domestic sector $j \in \{h, cb, pb\}$:

$$\begin{aligned} s_{h,it} &= \frac{\text{Pension fund assets}_{it} + \text{Insurance assets}_{it}}{\text{GDP}_{it}} && \rightarrow S_h \text{ (non-bank capacity)} \\ s_{pb,it} &= \frac{\text{Bank deposit funding}_{it}}{\text{GDP}_{it}} && \rightarrow S_{pb} \text{ (private-bank size)} \\ s_{cb,it} &= \frac{\text{Central bank assets}_{it}}{\text{GDP}_{it}} && \rightarrow S_{cb} \text{ (central-bank size)}. \end{aligned}$$

No country-specific size proxy is available for foreign investors, since foreign creditors are global balance sheets rather than a local sector we can size in GFDD. We therefore run *no first-stage regression for foreigners* and rely on (17) to recover $d_{f,it}$.

5.2 Empirical strategy: two-stage estimator

Threshold. $\bar{D} = 102.7\%$ of GDP, estimated from the data in §6.3: banking’s marginal absorption accelerates sharply once D_{it} crosses this level (Section 2).

Stage 1—structural sector size $\hat{\delta}_{ij}$. For each country i and sector $j \in \{h, cb, pb\}$, regress sector share on total debt with country and year fixed effects, allowing a kink at $\bar{D}$:

$$s_{ijt} = \delta_{ij} + \delta_{jt} + \alpha_{1j}D_{it} + \alpha_{2j}\mathbf{1}\{D_{it} \geq \bar{D}\}(D_{it} - \bar{D}) + u_{ijt}. \quad (18)$$

We read $\hat{\delta}_{ij}$, the country fixed effect, as the part of country i ’s sector- j size that does not move with time, after stripping out variation explained by D_{it} and the kink.

Stage 2—debt allocation. The effective slope is a product of a baseline rate and a size scaling, with a kink at $\bar{D}$. In its original factored form,

$$d_{it}^j = \underbrace{\beta_j(1 + \gamma_j\hat{\delta}_{ij})}_{\text{baseline slope}} D_{it} + \mathbf{1}_{it} \underbrace{\eta_j(1 + \rho_j\hat{\delta}_{ij})}_{\text{post-threshold slope change}} (D_{it} - \bar{D}) + \varepsilon_{ijt}. \quad (19)$$

Expanding the two products gives the estimating equation reported in the tables:

$$d_{it}^j = \beta_j D_{it} + \theta_j(\hat{\delta}_{ij} \cdot D_{it}) + \mu_j \mathbf{1}_{it}(D_{it} - \bar{D}) + \nu_j \mathbf{1}_{it}(\hat{\delta}_{ij} \cdot (D_{it} - \bar{D})) + \varepsilon_{ijt}, \quad (20)$$

with $\theta_j = \beta_j \gamma_j$, $\mu_j = \eta_j$, and $\nu_j = \eta_j \rho_j$.

- β_j : baseline slope—sector j ’s absorption per unit of D .
- θ_j : *size-scaling*—larger $\hat{\delta}_{ij}$ implies steeper absorption.
- μ_j, ν_j : post-threshold slope change and its size-scaling.

Inference. Standard errors throughout the empirical section are two-way cluster-robust at the country and year level, following Cameron, Gelbach, and Miller (2011). This

is more conservative than one-way country clustering and allows for arbitrary within-country serial correlation *and* arbitrary within-year cross-country correlation in the residuals.

5.3 Estimating the high-debt threshold

We estimate the threshold $\bar{D}$ from the data using a Hansen-style threshold regression of total banking-sector holdings, $db_{it} \equiv d_{cb,it} + d_{pb,it}$, on the level of total debt D_{it} :

$$db_{it} = \alpha_1 D_{it} \mathbf{1}\{D_{it} < \bar{D}\} + \alpha_2 D_{it} \mathbf{1}\{D_{it} \geq \bar{D}\} + \xi_{it},$$

with $\bar{D}$ chosen by least squares on a grid of candidate values. Column (1) of Table 1 includes country fixed effects, identifying $\bar{D}$ from within-country variation in debt levels; column (2) is the pooled cross-section.

Table 1 Banking-sector threshold regression

	(1) Country FE	(2) Pooled
Debt threshold $\hat{\bar{D}}$	102.7%	106.2%
<i>Debt below threshold</i>		
D	0.274*** (18.41)	0.254*** (22.49)
cons	-1.037 (-0.85)	0.902 (1.40)
<i>Debt above threshold</i>		
D	0.964*** (30.39)	1.048*** (37.23)
cons	-76.90*** (-19.14)	-89.10*** (-23.15)
Obs.	780	780

z statistics in parentheses. * $p < .05$, ** $p < .01$, *** $p < .001$. Dependent variable: $db_{it} = d_{cb,it} + d_{pb,it}$. Threshold variable and region regressor: D_{it} . Country fixed-effect estimates suppressed in column (1).

5.4 First-stage results: structural sector sizes

Table 2 reports estimates of (18).

The estimates split cleanly along the model's two-versus-one taxonomy. For non-banks, neither α_{1h} nor α_{2h} is statistically distinguishable from zero, so the size of the non-bank sector does not move with total debt either below or above the threshold. This is what we would expect under a *hard* capacity $\bar{\omega}S_h$ whose level evolves country

Table 2 First stage: structural sector sizes

	(1) Non-bank s_{iht}	(2) Central bank s_{icbt}	(3) Private bank s_{ipbt}
$\alpha_{1j}: D_{it}$	-0.198 (0.217)	0.023 (0.040)	-0.309 (0.258)
$\alpha_{2j}: \text{highD} (D_{it} - \bar{D})$	-0.219 (0.183)	0.542*** (0.158)	0.882** (0.402)
Obs.	547	718	732
Groups	31	32	32
R^2 (within)	0.436	0.611	0.200
Year / Country FE	Yes	Yes	Yes

Two-way cluster-robust standard errors (country and year) in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

by country but does not respond to the sovereign’s issuance choices. The two banking-sector regressions tell the opposite story: α_{2j} is positive and significant in both the central-bank and private-bank columns, so banking-sector size *grows* once D_{it} crosses $\bar{D}$. That growth is the empirical counterpart of banking’s no-hard-ceiling property in the model — when non-banks and foreigners saturate, the planner channels marginal debt onto the banking system and the banking system itself expands to absorb it.

5.5 Second-stage results: debt allocation

Table 3 reports estimates of (20).

The Stage 2 estimates line up with three of the four model predictions and refine the fourth. The baseline slopes β_j are positive and highly significant across all three sectors, confirming that every domestic creditor absorbs some share of marginal issuance — the empirical counterpart of Proposition 1. The non-bank slope scales further with structural size: $\theta_h > 0$ at conventional significance, so countries with intrinsically larger non-bank sectors absorb a larger fraction of any given D , exactly the size-scaling channel of Proposition 2. Above the threshold the picture becomes one of banking dominance. The combined post-threshold slope $\mu_{cb} + \mu_{pb} \approx 0.82$ implies that, once D_{it} crosses $\bar{D}$, banking absorbs roughly 82 percentage points more of each additional unit of debt than below; the private-bank component ($\mu_{pb} = 0.54$, $p = 0.03$) is significant at conventional levels and the central-bank component ($\mu_{cb} = 0.28$, $p = 0.08$) at the 10% level under two-way clustering. This is the banking-takeover prediction of Section 4 in quantitative form.

Table 3 Second stage: debt allocation by sector

	Non-bank (d_h)	Central bank (d_{cb})	Private bank (d_{pb})
β_j : D_{it}	0.315*** (0.039)	0.057*** (0.014)	0.194*** (0.019)
θ_j : $\hat{\delta}_{ij} \cdot D_{it}$	0.0018*** (0.0005)	0.0127*** (0.0043)	-0.0002 (0.0007)
μ_j : threshold slope	-0.037 (0.165)	0.281* (0.159)	0.537** (0.243)
ν_j : threshold $\times \hat{\delta}$	-0.003 (0.003)	-0.063*** (0.022)	-0.012 (0.009)
Obs.	753	780	780
F	75.76	33.13	137.00
R^2	0.802	0.713	0.821

Two-way cluster-robust standard errors (country and year) in parentheses. Regressions omit a constant by construction. F from joint test on all four regressors. R^2 is uncentered (no constant). * $p < .10$, ** $p < .05$, *** $p < .01$.

Foreign debt residual. Using the fitted domestic components from (20), the implied foreign residual tracks the data: correlation ≈ 0.78 , slope ≈ 0.74 , $R^2 \approx 0.61$, MAE ≈ 8.8 .

5.6 Concavity in debt absorption

Specification. For sector $j \in \{h, f\}$,

$$d_{it}^j = \beta_j D_{it} + \kappa_j D_{it}^2 + \theta_j (\hat{\delta}_h \cdot D_{it}) + \varepsilon_{it}, \quad (21)$$

where θ_j is included only for the non-bank sector. Pooled and country-FE estimates are reported separately, with and without $\hat{\delta}_h \cdot D$.

Table 4 reports estimates of (21).

The two capped sectors show concavity, but at different margins. For non-banks the curvature term κ_h is negative and significant in the country-FE specification but moves toward zero once we condition on the size interaction $\hat{\delta}_h \cdot D$, so the within-country slowdown is driven by country-specific non-bank capacity S_h rather than by an absolute level of debt. For foreigners, by contrast, $\kappa_f < 0$ shows up only in the pooled specification and not within country, which is exactly the signature of a *common* foreign ceiling $\bar{D}_f$ that is shared across the panel: the curvature is identified by cross-country variation in how close each economy sits to the same underlying limit. Taken together these two findings deliver Proposition 3: the marginal absorption shares of non-banks and

Table 4 Concavity in debt absorption

	Non-bank (d_h)				Foreign (d_f)	
	Spec C: no $\hat{\delta}_h$		Spec B: with $\hat{\delta}_h \cdot D$		No $\hat{\delta}_h \cdot D$	
	Pooled	FE	Pooled	FE	Pooled	FE
$\beta(D)$	0.343***	0.402***	0.359***	0.299*	0.387***	0.392***
$\theta(\hat{\delta}_h \cdot D)$	—	—	0.0022***	0.0011*	—	—
$\kappa(D^2)$	−0.0003	−0.0016***	−0.0010	−0.0009	−0.0014***	−0.0009
N	780	780	701	701	747	747
R^2	0.735	0.381	0.801	0.381	0.773	0.454

Standard errors omitted for space. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

foreigners both decline as D grows, but the source of concavity differs by sector — country-specific S_h for non-banks, a common $\bar{D}_f$ for foreigners.

Robustness appendix. Appendix B reports matched-sample and sparse-country variants of the Stage 2 specifications in Tables 1–3. The headline size-scaling coefficients and the magnitude of the banking takeover ($\mu_{cb} + \mu_{pb}$) are stable across these designs.

6 Conclusion

This paper studies how governments allocate debt across creditor sectors. The starting point is a simple empirical puzzle: countries with similar levels of public debt often have very different creditor mixes. Some rely heavily on domestic non-bank investors, others on foreign investors, and others on the banking system. At the same time, the data show a sharp change at high debt levels: once debt rises to around 100% of GDP, the banking sector begins to absorb a much larger share of additional government debt.

To explain these patterns, the paper develops a cost-minimizing model of sovereign debt allocation. The government must place a given stock of debt and chooses how much to allocate to domestic non-banks, foreign investors, and the banking sector. The key distinction across sectors is their effective capacity. Domestic non-bank capacity is country-specific and depends on the depth of the local financial system. Foreign demand is limited by a common exposure ceiling. The banking sector has no hard ceiling, but greater reliance on banks carries increasing quasi-fiscal and financial-stability costs. This structure generates a pecking-order logic: at low and moderate debt levels, non-banks and foreigners absorb much of the debt; as these sectors become saturated, marginal debt absorption shifts toward the banking system.

The empirical results support the main predictions of the model. Using a panel of 32 advanced economies, the paper finds that countries with larger structural non-bank

sectors absorb more government debt through non-bank investors at a given debt level. This size-scaling result is consistent with the model's prediction that domestic financial structure shapes the creditor composition of public debt. The results also show two distinct forms of concavity. Non-bank absorption becomes concave within countries, consistent with country-specific domestic capacity constraints. Foreign absorption is instead concave mainly across countries, consistent with a common foreign-investor ceiling. Finally, the estimated high-debt threshold is around 103% of GDP. Above this threshold, the combined marginal absorption of central banks and private banks rises sharply, indicating a banking-sector takeover of marginal government debt.

These findings suggest that the fiscal consequences of high public debt depend not only on the level of debt, but also on the structure of the investor base. A country with a deep domestic non-bank sector can place more debt outside the banking system and may therefore face a different form of fiscal pressure than a country with a thinner non-bank investor base. Conversely, when non-bank and foreign demand become saturated, additional debt is more likely to migrate toward the banking sector, strengthening the sovereign-bank nexus and raising the risk of financial repression or quasi-fiscal costs.

A Proofs of model propositions

This appendix collects proofs of Propositions 1, 2, and 3 stated in Section 4. Throughout, $H_j \equiv \partial MC_j / \partial d_j$ denotes the slope of sector j 's marginal-cost schedule, with closed forms given in equations (13)–(15); the structural property used at every step is $H_j > 0$ in each sector.

Proof of Proposition 1. Suppose, for contradiction, that D rises but d_h^* falls. Because $H_h > 0$ strictly, MC_h falls, so the new equilibrium shadow rate λ is lower. The first-order condition (7) then implies that MC_f and $\Psi'(d_b)$ also fall, so d_f^* and d_b^* fall as well (since $H_f, H_b > 0$). But then $d_h^* + d_f^* + d_b^* = D$ has declined, contradicting the premise that D rose. The same argument applies to d_f^* and d_b^* by symmetry. Strict positivity of every H_j delivers strict inequality at any interior allocation. $\square$

Proof of Proposition 2. Differentiate the FOC (7) and the budget constraint (1) with respect to S_h , holding D fixed. Write $K_h \equiv \partial MC_h / \partial S_h$ at fixed d_h . Then

$$H_h dd_h + K_h dS_h = d\lambda, \quad H_f dd_f = d\lambda, \quad H_b dd_b = d\lambda, \quad dd_h + dd_f + dd_b = 0.$$

Using the last three equations to eliminate $dd_f, dd_b, d\lambda$ in terms of dd_h yields

$$\left(H_h + \frac{H_f H_b}{H_f + H_b} \right) dd_h = -K_h dS_h.$$

Because MC_h depends on d_h only through the share $x \equiv d_h / S_h$, the chain rule gives $K_h = -(\partial MC_h / \partial x) d_h / S_h^2 = -H_h d_h / S_h < 0$ whenever $d_h > 0$. The bracket multiplying dd_h is strictly positive, hence $\partial d_h^* / \partial S_h > 0$. $\square$

Proof of Proposition 3. Let $m \equiv \lambda - \bar{r}_h > 0$. Differentiating the closed-form solutions (10)–(12) in λ ,

$$A_h(\lambda) \equiv \frac{dd_h^*}{d\lambda} = \frac{S_h \bar{\omega}}{2\gamma_h} (1+m/\gamma_h)^{-3/2}, \quad A_f(\lambda) \equiv \frac{dd_f^*}{d\lambda} = \frac{\bar{D}_f}{2\gamma_f} (1+m/\gamma_f)^{-3/2}, \quad \frac{dd_b^*}{d\lambda} = \frac{1}{\psi}.$$

Because $D'(\lambda) = A_h + A_f + 1/\psi > 0$, the change of variables $\lambda \mapsto D(\lambda)$ is a strictly increasing C^1 -diffeomorphism, with

$$\frac{dd_h^*}{dD} = \frac{1}{1 + A_f/A_h + 1/(\psi A_h)}. \quad (22)$$

Under the symmetric calibration $\bar{r}_h = \bar{r}_f$ and $\gamma_h = \gamma_f \equiv \gamma$, the λ -dependent factors in A_h and A_f are identical, so $A_f/A_h = \bar{D}_f/(S_h \bar{\omega})$ is constant in λ . The denominator of (22) can therefore be written

$$\varphi(\lambda) = \left(1 + \frac{\bar{D}_f}{S_h \bar{\omega}}\right) + \frac{2\gamma}{\psi S_h \bar{\omega}} (1 + m/\gamma)^{3/2},$$

which is strictly increasing in λ on $[\bar{r}_h, \infty)$, so $dd_h^*/dD = 1/\varphi(\lambda)$ is strictly decreasing in λ . Since $D(\lambda)$ is strictly increasing, dd_h^*/dD is also strictly decreasing in D , which is equivalent to strict concavity of $d_h^*(D)$. Replacing the roles of A_h and A_f in the same argument delivers strict concavity of $d_f^*(D)$. $\square$

B Robustness of the two-stage estimator

The Stage 1 / Stage 2 design recovers a country-level structural-size effect $\hat{\delta}_{ij}$ from a fixed-effects regression of sector size on debt (Equation (18)), and uses $\hat{\delta}_{ij}$ as a country-constant regressor in Stage 2 (Equation (20)). Two features of this design merit robustness checks.

Sample mismatch. The Stage 1 sample uses GFDD sector-size data, which is sparser than the Arslanalp–Tsuda holdings data used in Stage 2. After Stage 1, $\hat{\delta}_{ij}$ is constructed as the country-level fixed effect and is propagated to every country-year of country i , including country-years where the underlying size variable was *not* observed. As a result, Stage 2 includes country-years that were not in the Stage 1 sample.

Sparsely-measured countries. $\hat{\delta}_{ij}$ is more precisely estimated for countries with many Stage 1 observations and noisier for countries with few. The per-country Stage 1 observation counts are: most countries contribute ≥ 11 non-bank size observations; the exceptions are *Cyprus* (0, dropped automatically because $\hat{\delta}_{i,h}$ is missing), *Luxembourg* (1) and *Malta* (4). For the central-bank size, only *Latvia* has fewer than 10 observations (9). For private-bank size, only *Canada* has fewer than 10 (8).

We address both concerns with two robustness specifications.

Matched sample. Stage 2 is restricted to country-years that were also in the Stage 1 sample for the corresponding sector, eliminating the propagated rows (only rows where $s_{ij,it}$ was directly observed are kept).

sparse-country drop. Stage 2 drops countries with fewer than k Stage 1 observations for the corresponding sector. We report $k \geq 5$ and $k \geq 10$.

All specifications use the Hansen threshold estimate $\hat{D} = 102.7\%$ of GDP (Section 5.3 and Table 1) and two-way cluster-robust standard errors at country and year. Robustness regressions are implemented jointly with the canonical Stage 2 in `Code/empirical_debt_composition/python/twoway_cluster_full.py`; the Stata analogue is `Code/empirical_debt_composition/do/twoway_cluster_full.do` (driven by `run_all.do`).

B.0.1 Robustness results

Headline coefficients (Tables 1–3) are reproduced in column (1) of each table below. Columns (2)–(4) report matched and sparse-country specifications. In every column, θ_{cb} rounds to 0.013 and stays significant two-way clustered; the headline non-bank size-scaling $\theta_h \in [0.002, 0.003]$ remains significant at least at the 1% level; and $\mu_{cb} + \mu_{pb}$ stays close to ~ 0.82 apart from matched samples where it softens modestly toward 0.78.

Table 5 Non-bank holdings (d_h): Stage 2 robustness

	(1) Headline	(2) Matched	(3a) $k \geq 5$	(3b) $k \geq 10$
$\beta_h: D_{it}$	0.315*** (0.039)	0.297*** (0.038)	0.314*** (0.041)	0.314*** (0.041)
$\theta_h: \hat{\delta}_h \cdot D_{it}$	0.0018*** (0.0005)	0.0021*** (0.001)	0.0019*** (0.001)	0.0019*** (0.001)
μ_h : threshold slope	−0.037 (0.165)	−0.055 (0.146)	−0.031 (0.169)	−0.031 (0.169)
ν_h : threshold $\times \hat{\delta}$	−0.003 (0.003)	−0.002 (0.003)	−0.003 (0.003)	−0.003 (0.003)
Obs.	753	547	709	709
Countries	31	31	29	29
R^2	0.802	0.818	0.804	0.804

Notes: Columns match `twoway_cluster_full`: headline, Stage 1-matched Stage 2, and sparse-country drops $k \geq 5$ / $k \geq 10$. R^2 is uncentered (Stage 2 has no intercept). Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (based on asymptotic cluster-robust z ratios). Columns (3a) and (3b) coincide numerically because the marginal countries below $k \geq 10$ for non-bank Stage 1 are already excluded by the $k \geq 5$ subset in this baseline.

Table 6 Central-bank holdings (d_{cb}): Stage 2 robustness

	(1) Headline	(2) Matched	(3a) $k \geq 5$	(3b) $k \geq 10$
$\beta_{cb}: D_{it}$	0.057*** (0.014)	0.060*** (0.014)	0.057*** (0.014)	0.057*** (0.014)
$\theta_{cb}: \hat{\delta}_{cb} \cdot D_{it}$	0.0127*** (0.0043)	0.0127** (0.0045)	0.0127*** (0.0044)	0.0128*** (0.0044)
μ_{cb} : threshold slope	0.281* (0.159)	0.274 (0.162)	0.281* (0.164)	0.283 (0.169)
ν_{cb} : threshold $\times \hat{\delta}$	-0.063*** (0.022)	-0.064** (0.026)	-0.063** (0.027)	-0.063** (0.023)
Obs.	780	718	780	762
Countries	32	32	32	31
R^2	0.713	0.716	0.713	0.714

Notes: Columns (1), (3a), and (3b) differ only mildly for columns (3a) versus (3b) except for the sample tightening to 762 observations when countries with < 10 private-bank size observations drop out of Stage 2's cross-country feasibility set.

Table 7 Private-bank holdings (d_{pb}): Stage 2 robustness

	(1) Headline	(2) Matched	(3a) $k \geq 5$	(3b) $k \geq 10$
$\beta_{pb}: D_{it}$	0.194*** (0.019)	0.191*** (0.019)	0.194*** (0.019)	0.194*** (0.020)
$\theta_{pb}: \hat{\delta}_{pb} \cdot D_{it}$	-0.0002 (0.0007)	-0.0002 (0.001)	-0.0002 (0.0007)	-0.0002 (0.001)
μ_{pb} : threshold slope	0.537** (0.243)	0.526** (0.223)	0.537** (0.250)	0.536** (0.249)
ν_{pb} : threshold $\times \hat{\delta}$	-0.012 (0.009)	-0.011 (0.008)	-0.012 (0.009)	-0.012 (0.009)
Obs.	780	732	780	759
Countries	32	32	32	31
R^2	0.821	0.823	0.821	0.817

Notes: As in Tables 2–3, Cyprus is unavailable for Stage 2 non-bank specifications; matched samples follow the tighter Stage 1 coverage for banks.

B.0.2 Banking takeover sum across specifications

Table 8 Combined banking-sector threshold slope $\mu_{cb} + \mu_{pb}$

Specification	$\mu_{cb} + \mu_{pb}$
(1) Headline (Tables 1–3)	0.82
(2) Matched sample	0.80
(3a) Sparse drop $k \geq 5$	0.82
(3b) Sparse drop $k \geq 10$	0.82

Summary. Two implementation concerns about the two-stage estimator do not materially affect the headline findings. (i) The *propagation* of $\hat{\delta}_{ij}$ to country-years not in the Stage 1 sample is innocuous: restricting Stage 2 to the matched sample leaves the size-scaling and threshold-slope coefficients within sampling variability of the headline estimates and, for non-banks, slightly strengthens θ_h . (ii) The presence of countries with very few Stage 1 observations (Luxembourg, Malta for non-banks; Latvia, Canada for banking-size variables) does not drive the size-scaling result: dropping these countries leaves the headline coefficients essentially unchanged. The banking-takeover sum $\mu_{cb} + \mu_{pb}$ is between 0.78 and 0.82 in every specification.

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